

Fixed-cost compatibility, balanced binary costs, and the forest boundary

Solution packet for arXiv:1608.04477

9 August 2026

Status. Likely-valid strengthened partial answer, pending human review. We give an exact fixed-cost compatibility criterion, a complete classification for binary marginal spaces, and the earlier costs-independent iff-forest classification.

1 Source question and scope

Bartz, Bauschke, and Wang study costs

$$c(x_1, \dots, x_N) = \sum_{1 \leq i < j \leq N} c_{ij}(x_i, x_j)$$

and ask on p. 12 of [1]:

For which cost functions of this form does c -cyclical monotonicity of a set Γ imply c_{ij} -cyclical monotonicity of its two-marginal projections? Given a cost, for which sets Γ does this implication hold?

Our discussion thus far raises the following natural, currently unsolved, questions: ① For which cost functions c of the form (8), c -cyclic monotonicity of a set Γ implies the $c_{i,j}$ -cyclic monotonicity of its $\Gamma_{i,j}$'s? ② Given a cost function c , for which sets Γ , c -cyclic monotonicity of a set Γ implies the $c_{i,j}$ -cyclic monotonicity of its $\Gamma_{i,j}$'s?

Figure 1: The two questions in the source paper [1, p. 12].

There are three levels to the answer. First, for an arbitrary fixed family of pair costs, we prove that the desired implication is equivalent to the existence, for every finitely supported family of one-marginals, of one joint coupling which is simultaneously optimal on every interacting pair. This is an exact criterion, but not yet an intrinsic classification of arbitrary cost functions. Second, when every X_i has two points, we turn the criterion into a complete and checkable answer: the signed graph of nonzero 2×2 cross differences must be balanced. Third, uniformly over all sets and all pair costs supported on a graph, the answer is exactly that the graph is a forest.

2 Definitions

Let $G = (V, E)$ be a finite simple graph, with $V = \{1, \dots, N\}$. For each $i \in V$, let X_i be a nonempty set. For every edge $e = \{i, j\} \in E$, let $c_e : X_i \times X_j \rightarrow \mathbb{R}$ be arbitrary, and set

$$c(x_1, \dots, x_N) = \sum_{e=\{i,j\} \in E} c_e(x_i, x_j). \quad (1)$$

For $\Gamma \subseteq \prod_i X_i$, write $\Gamma_e = \{(x_i, x_j) : x \in \Gamma\}$.

The set Γ is *c-cyclically monotone* if, for every $n \geq 2$, points $x^1, \dots, x^n \in \Gamma$, and permutations $\sigma_1, \dots, \sigma_n \in S_n$,

$$\sum_{r=1}^n c(x_1^{\sigma_1(r)}, \dots, x_N^{\sigma_N(r)}) \leq \sum_{r=1}^n c(x^r). \quad (2)$$

The analogous two-coordinate definition is used for Γ_e and c_e .

Say that G has the *universal projection property* if, for every choice above,

$$\Gamma \text{ is } c\text{-cyclically monotone} \implies \Gamma_e \text{ is } c_e\text{-cyclically monotone for every } e \in E.$$

3 An exact fixed-cost compatibility criterion

We first keep the sets X_i and the numerical pair costs c_e fixed. For finitely supported probability measures μ_i on X_i , write $\Pi(\mu_1, \dots, \mu_N)$ for their joint couplings. We say that the family $(c_e)_{e \in E}$ has the *edgewise optimal compatibility property* if, for every such $(\mu_i)_i$, there is a $\pi \in \Pi(\mu_1, \dots, \mu_N)$ for which

$$\pi_e \in \operatorname{argmax}_{\gamma \in \Pi(\mu_i, \mu_j)} \int c_e d\gamma \quad \text{for every } e = \{i, j\} \in E. \quad (3)$$

Only finite transportation polytopes occur in this definition, so no topological or integrability hypotheses are needed.

Lemma 1 (Finite optimality bridge). *Let d be any real-valued cost on a finite product of arbitrary sets.*

- (a) *If Λ is d -cyclically monotone, every finitely supported probability measure carried by Λ maximizes $\int d$ among couplings of its one-coordinate marginals.*
- (b) *The support of every finitely supported maximizing coupling is d -cyclically monotone.*

Proof. For (a), suppose that a finitely supported π carried by Λ has a strictly better competitor ρ . Restrict to the finite product of the supports of the marginals. Pairs (p, q) of nonnegative arrays such that p is carried by Λ , p and q have the same one-marginals, and both have mass one form a polytope defined by integer equalities and inequalities. Rational points are dense in every nonempty face of such a rational polytope. Since the cost gap is strict, there is a rational pair (p, q) with the same positive gap.

Clear denominators. List the atoms of p , with their integer multiplicities, as $x^1, \dots, x^m \in \Lambda$, and list the atoms of q in the same way as $y^1, \dots, y^m$. Equality of each one-marginal says that, for every coordinate i , the list $(y_i^r)_r$ is a permutation of $(x_i^r)_r$. Thus q is one of the coordinatewise rearrangements forbidden by d -cyclical monotonicity, a contradiction.

For (b), if points $x^1, \dots, x^m$ in the support and coordinate permutations produced a strict gain, subtract a sufficiently small equal mass from the selected support points (counting repetitions) and add it to the permuted points. This preserves all one-marginals and strictly raises the objective, contradicting optimality. $\square$

Theorem 2 (Fixed-cost equivalence). *For fixed X_i and fixed pair costs $(c_e)_{e \in E}$, the following are equivalent.*

- (i) *For every $\Gamma \subseteq \prod_i X_i$, c -cyclical monotonicity of Γ implies c_e -cyclical monotonicity of Γ_e for every edge.*
- (ii) *The family $(c_e)_{e \in E}$ has the edgewise optimal compatibility property (3).*

Proof. Assume (i), fix finitely supported marginals, and let π maximize the total cost $c = \sum_e c_e$. By Lemma 1(b), $\Gamma = \text{supp } \pi$ is c -cyclically monotone. Hence every Γ_e is c_e -cyclically monotone, and Lemma 1(a), now in two coordinates, shows that every π_e is pairwise optimal. Thus π satisfies (3).

Conversely, assume (ii). Let Γ be c -cyclically monotone, choose finitely many lifts $x^1, \dots, x^m \in \Gamma$, and let

$$\pi^0 = \frac{1}{m} \sum_{r=1}^m \delta_{x^r}.$$

By Lemma 1(a), π^0 is globally optimal for its one-marginals. By (ii), those marginals admit a coupling $\hat{\pi}$ all of whose edge marginals are pairwise optimal. If M_e denotes the pairwise optimal value, then

$$\sum_e \int c_e d\pi_e^0 \geq \sum_e \int c_e d\hat{\pi}_e = \sum_e M_e \geq \sum_e \int c_e d\pi_e^0.$$

Every inequality is therefore equality, and each π_e^0 is pairwise optimal. Comparing it with any coordinatewise rearrangement of the chosen edge pairs gives the c_e -cyclical inequality. The edge, the lifts, and the rearrangement were arbitrary, proving (i). $\square$

This theorem is a full equivalence for a fixed cost family. Its value is that it replaces a quantifier over all subsets Γ and all cycle lengths by a standard marginal-compatibility question. For general finite alphabets, however, condition (3) is still a substantial compatibility problem rather than an intrinsic normal form for the matrices c_e .

4 Complete classification for binary spaces

Now take $X_i = \{0, 1\}$ for every vertex. Define the cross difference

$$\Delta_e = c_e(0, 0) + c_e(1, 1) - c_e(0, 1) - c_e(1, 0). \quad (4)$$

Modulo one-coordinate terms, $c_e(s, t) = \Delta_e st$. Let H be the graph of edges with $\Delta_e \neq 0$, and give such an edge the sign $\varepsilon_e = \text{sgn}(\Delta_e)$. A signed graph is *balanced* if the product of the edge signs around every cycle is $+1$.

Theorem 3 (Binary iff-balance theorem). *For binary marginal spaces and arbitrary real pair costs, the following are equivalent.*

- (i) *Global c -cyclical monotonicity forces c_e -cyclical monotonicity of every edge projection, for every set $\Gamma \subseteq \{0, 1\}^V$.*
- (ii) *The signed graph (H, ε) is balanced.*
- (iii) *There are vertex signs $\eta_i \in \{-1, 1\}$ such that $\varepsilon_{\{i, j\}} = \eta_i \eta_j$ on every edge of H .*

If the signed graph is unbalanced, failure in (i) already has a two-point witness of the form $\Gamma = \{a, \mathbf{1} - a\}$.

Proof. The equivalence of (ii) and (iii) is the elementary switching characterization of balanced signed graphs: define η_i as the product of edge signs along a path from a chosen root; this is path-independent exactly when every cycle has positive sign product.

Assume (iii). At every vertex with $\eta_i = -1$, relabel the binary coordinate by $s \mapsto 1 - s$. Relabeling one endpoint reverses the sign of the cross difference, so all transformed nonzero cross differences are positive. Given arbitrary Bernoulli one-marginals in the transformed labels, take one uniform random variable U and set

$$Y_i = \mathbf{1}_{\{U \leq p_i\}}.$$

For every edge, this comonotone coupling maximizes $\mathbb{E}[Y_i Y_j]$, and hence maximizes the transformed pair cost; edges with zero cross difference are indifferent. Undoing the relabelings gives one joint coupling simultaneously optimal on every edge. Theorem 2 proves (i).

Now suppose that the signed graph is unbalanced. Give every coordinate the Bernoulli(1/2) marginal. On a positive edge the unique pairwise optimum has equal bits almost surely; on a negative edge it has opposite bits almost surely. These requirements are inconsistent around an unbalanced cycle, so edgewise optimal compatibility fails and Theorem 2 already disproves (i).

For the promised explicit form, choose $a \in \{0, 1\}^V$ maximizing the total cost of

$$\pi_a = \frac{1}{2}(\delta_a + \delta_{\mathbf{1}-a})$$

among all such complementary pairs. Any joint law with Bernoulli(1/2) marginals may be symmetrized under $x \mapsto \mathbf{1} - x$ without changing any pair-cost expectation. The symmetrized law is a mixture of the π_a 's. Hence π_a is in fact a globally optimal joint coupling. Lemma 1(b) makes its support $\Gamma = \{a, \mathbf{1} - a\}$ globally c -cyclically monotone. No assignment a satisfies all signed edge relations on an unbalanced cycle, so some edge has the wrong relation; its two-point projection violates the corresponding c_e -cyclical inequality. $\square$

Corollary 4 (Consistently supermodular costs). *Suppose each X_i is totally ordered and, with respect to these vertex orders, every c_e is supermodular:*

$$c_e(x, y) + c_e(x', y') \geq c_e(x, y') + c_e(x', y) \quad (x \leq x', y \leq y').$$

Then the fixed family $(c_e)_e$ has the projection property in Theorem 2(i).

Proof. For finitely supported marginals, use a common uniform random variable and the quantile function of every marginal. The resulting joint coupling is comonotone on every edge. The elementary adjacent-swap rearrangement argument shows that each such pair coupling maximizes every supermodular edge cost. Apply Theorem 2. $\square$

The binary theorem says that the common-order mechanism is also necessary on two-point spaces: vertexwise label reversals make every edge supermodular exactly when the cross-difference signed graph is balanced.

5 The costs-independent graph boundary

An edge of a forest is a bridge. Delete it and use one permutation on all coordinates on one side and a second permutation on all coordinates on the other side. The sum associated with every internal edge is unchanged by a common reindexing. The deleted edge is the only term left in the global cyclical inequality, so its two-marginal inequality follows.

A cycle is precisely the obstruction. On binary coordinates, compare the two aligned points 0^V and 1^V . Under arbitrary coordinate permutations, each row determines a graph cut. Choose one cycle edge with weight -1 and all other cycle edges with weight $+1$. Every cut meets a cycle in an even number of edges, so every cut has nonnegative total weight; this proves all global cyclical inequalities at once. The negative edge itself violates its two-marginal inequality.

5.1 Exact characterization

Theorem 5 (Universal iff-forest theorem). *For a finite simple graph G , the following are equivalent.*

- (i) G is a forest.
- (ii) G has the universal projection property.

If G contains a cycle, failure in (ii) already occurs with $X_i = \{0, 1\}$ and a two-point set Γ .

Proof. Assume first that G is a forest. Fix an edge $e = \{a, b\}$. Deleting e splits its tree component into vertex sets A and B , with $a \in A$ and $b \in B$; all other connected components of G are irrelevant.

Take n points $x^1, \dots, x^n \in \Gamma$ and arbitrary $\alpha, \beta \in S_n$. In (2), set

$$\sigma_i = \alpha \quad (i \in A), \quad \sigma_i = \beta \quad (i \in B),$$

and use the identity permutation on every other component. If an edge has both endpoints in A (or both in B), its total contribution is unchanged, because a common permutation merely reindexes the sum over r . The same is true in every other component. Since e is the unique edge between A and B , cancellation in (2) leaves exactly

$$\sum_{r=1}^n c_e(x_a^{\alpha(r)}, x_b^{\beta(r)}) \leq \sum_{r=1}^n c_e(x_a^r, x_b^r).$$

This is the required c_e -cyclical inequality. Since e, n, α, β were arbitrary, every Γ_e is c_e -cyclically monotone.

Conversely, suppose that G contains a simple cycle C , and fix an edge $e_0 \in C$. Let $X_i = \{0, 1\}$ for all i and

$$\Gamma = \{0^V, 1^V\}.$$

Assign edge weights

$$w_e = \begin{cases} -1, & e = e_0, \\ 1, & e \in C \setminus \{e_0\}, \\ 0, & e \notin C, \end{cases} \quad c_e(s, t) = w_e \mathbf{1}_{\{s=t\}}.$$

We first prove that Γ is c -cyclically monotone. Choose any n points from Γ , allowing repetitions, and encode them by bits $b_1, \dots, b_n$: the r th point is b_r^V . For arbitrary coordinate permutations σ_i , define for each row

$$S_r = \{i \in V : b_{\sigma_i(r)} = 1\}.$$

The original coordinates agree across every edge. Hence the difference between the original and permuted total costs is

$$\sum_{r=1}^n c(b_r^V) - \sum_{r=1}^n c(b_{\sigma_1(r)}, \dots, b_{\sigma_N(r)}) = \sum_{r=1}^n \sum_{e \in \delta(S_r)} w_e, \quad (5)$$

where $\delta(S)$ is the cut consisting of edges with exactly one endpoint in S .

Every cut intersects a cycle in an even number of edges. If e_0 is not in $\delta(S)$, its cut weight is nonnegative. If e_0 is in the cut, at least one other edge of C is also in the cut, and therefore

$$\sum_{e \in \delta(S)} w_e = -1 + \#((C \setminus \{e_0\}) \cap \delta(S)) \geq 0.$$

Thus the right side of (5) is nonnegative for all choices, which is exactly (2).

On the other hand, $\Gamma_{e_0} = \{(0, 0), (1, 1)\}$ is not c_{e_0} -cyclically monotone. For the two points and the permutation that swaps the second coordinate, the original total is -2 while the permuted total is 0 , contradicting the required inequality $0 \leq -2$. Hence a graph containing a cycle does not have the universal projection property. $\square$

Corollary 6 (Removing inessential pair terms). *Suppose a pair term has the form $c_{ij}(x_i, x_j) = a_i(x_i) + a_j(x_j)$. It may be removed without changing any cyclical inequality. Consequently, if the graph of nonseparable pair terms is a forest, global cyclical monotonicity forces cyclical monotonicity of every remaining two-marginal projection.*

Proof. Under coordinate permutations, the total of a one-coordinate function is unchanged. Remove all such terms and apply Theorem 5. $\square$

6 Verification notes, limitations, and novelty

Lemma 1 explicitly handles the point at which multi-marginal arguments can go wrong: a finite multi-stochastic array need not itself be a permutation array. Repetition and rationalization are what turn an arbitrary strict transport improvement into one forbidden coordinatewise rearrangement. Thus Theorem 2 does not assume a false multi-marginal Birkhoff theorem.

The binary classification was also checked computationally on the triangle. The verification script enumerates all 256 subsets of $\{0, 1\}^3$ for each of the eight nonzero sign patterns. Finite LP splitting tests found the universal projection property for exactly the four balanced patterns and produced two-point witnesses for all four unbalanced patterns. This is only a sanity check; the proof of Theorem 3 is symbolic and works on every finite graph with arbitrary edge magnitudes and zero cross differences.

Theorem 2 is an exact answer in terms of simultaneous pairwise optimizer compatibility, and Theorem 3 is a complete intrinsic answer for two-point spaces. What remains open here is an intrinsic classification of arbitrary pair-cost matrices or functions satisfying (3). We also do not classify the individual sets Γ requested in the source paper’s second question.

A bounded search through 9 August 2026 used the exact source question and the phrases *simultaneously optimal pairwise couplings*, *marginal compatibility*, *balanced signed graph*, *binary*, and *multi-marginal cyclical monotonicity*. No exact statement of Theorems 2 or 3 was found. There is important adjacent work. Haasler, Ringh, Chen, and Karlsson [2] formulate discrete tree-structured multi-marginal transport through coupled pairwise problems and exhibit a triangle obstruction for cyclic graphs. The extremal-dependence literature reviews simultaneous comonotone and pairwise countermonotone couplings and their compatibility restrictions [4, 5]. Harary’s classical balance theorem [3] supplies the signed-graph terminology and the switching equivalence used in the binary proof. None of these sources found in the bounded search states the present cyclical-monotonicity characterizations. Novelty confidence remains *moderate*, not high.

Human-review recommendation. Verify the sign convention in (2), the cancellation after deleting a forest edge, the rational-polytope step in Lemma 1, and the complement symmetrization in Theorem 3. Then compare the fixed-cost equivalence against the marginal-polytope and extremal-dependence literature, and the forest statement against Remark 3 of [2], before any novelty claim.

References

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